



第四单元 化学实验与应用 (Unit 4 Chemistry Experiments and Applications)

第一章 实验室安全与仪器使用 (Chapter 1 Laboratory Safety and Instrument Operation)

1. 关于下列仪器使用的说法错误的是 (Which of the following statements about the use of the following instruments is incorrect?) ()



- A. 仪器①使用前, 需要检查两处是否漏水 (Before using instrument ①, it is necessary to check whether there is water leakage at two positions.)
- B. 仪器②可用于测定硫酸铜晶体中的结晶水含量 (Instrument ② can be used to determine the content of crystal water in copper sulfate crystals.)
- C. 仪器③和④可用于配制一定质量分数的 CuSO_4 溶液 (Instruments ③ and ④ can be used to prepare CuSO_4 solution with a certain mass fraction.)
- D. 仪器⑤可用于移取 $0.1 \text{ mol} \cdot \text{L}^{-1} \text{Na}_2\text{Cr}_2\text{O}_7$ 溶液 (Instrument ⑤ can be used to transfer $0.1 \text{ mol} \cdot \text{L}^{-1} \text{Na}_2\text{Cr}_2\text{O}_7$ solution.)

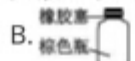
答案 (Answer): C

解析 (Analysis): 分液漏斗使用前需要检验下方的活塞和上方的玻璃塞是否漏水, A 正确; 测定硫酸铜晶体中的结晶水需要在坩埚中加热, B 正确; 配制一定质量分数的 CuSO_4 溶液, 需要天平、烧杯、玻璃棒和量筒, 不需要容量瓶, C 错误; $\text{Na}_2\text{Cr}_2\text{O}_7$ 溶液具有强氧化性, 故选择酸式滴定管移取, D 正确。 (Before using a separatory funnel, it is necessary to check whether the piston at the bottom and the glass stopper at the top leak water, so A is correct. Heating in a crucible is required to determine the crystal water in copper sulfate crystals, so B is correct. Preparing CuSO_4 solution with a certain mass fraction requires a balance, beaker, glass rod and measuring cylinder, and no volumetric flask is needed, so C is incorrect. $\text{Na}_2\text{Cr}_2\text{O}_7$ solution has strong oxidizing property, so an acid burette is selected for transfer, so D is correct.)

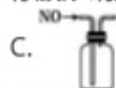
2. 下列有关实验操作或仪器的使用正确的是 (Which of the following experimental operations or instrument uses is correct?) ()



A. 分液时取出下层液体 (Take out the lower layer liquid during liquid separation.)



B. 存放浓硝酸 (Store concentrated nitric acid.)



C. 收集 NO 气体 (Collect nitric oxide gas.)



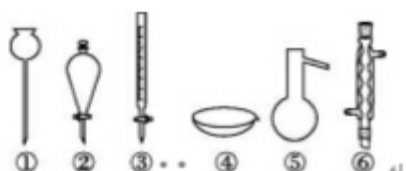
D. 吸收尾气中的 SO_2 (Absorb SO_2 in the tail gas.)

答案 (Answer): D



解析 (Analysis): 分液时下层液体从分液漏斗的下口放出, 上层液体从上口倒出, A 项错误; 浓硝酸具有强氧化性, 能氧化橡胶塞使其老化, 故应使用玻璃塞, B 项错误; NO 与空气中的氧气反应且 NO 密度与空气相近, 故应用排水法收集, C 项错误; D 项操作正确。(During liquid separation, the lower layer liquid is discharged from the lower opening of the separatory funnel, and the upper layer liquid is poured out from the upper opening, so A is incorrect. Concentrated nitric acid has strong oxidizing property and can oxidize the rubber stopper to cause aging, so a glass stopper should be used, so B is incorrect. Nitric oxide reacts with oxygen in the air and has a similar density to air, so it should be collected by water displacement method, so C is incorrect. Operation D is correct.)

3.关于下列仪器使用的说法正确的是 (Which of the following statements about the use of the following instruments is correct?) ()

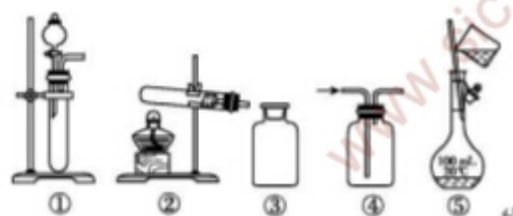


- A. 仪器①和②可用于物质分离 (Instruments ① and ② can be used for substance separation.)
- B. 仪器②和③使用时需要检查是否漏水 (Instruments ② and ③ need to be checked for water leakage before use.)
- C. 仪器④和⑤可用酒精灯直接加热 (Instruments ④ and ⑤ can be heated directly with an alcohol lamp.)
- D. 仪器③和⑥使用前需要烘干 (Instruments ③ and ⑥ need to be dried before use.)

答案 (Answer):B

解析 (Analysis): ①为长颈漏斗, 用于向容器中添加液体, 不能用于分离物质; ②为分液漏斗, 可用于分离互不相溶的液体, A 错误。一般带瓶塞或者旋塞的仪器, 为防止使用过程中漏水, 使用前需要检查是否漏水, B 正确。蒸发皿可以用酒精灯直接加热, 蒸馏烧瓶应垫陶土网加热, C 错误。酸式滴定管使用前不需要烘干, D 错误。(① is a long neck funnel, which is used to add liquid to the container and cannot be used for substance separation; ② is a separatory funnel, which can be used to separate immiscible liquids, so A is incorrect. Generally, instruments with stoppers or stopcocks need to be checked for water leakage before use to avoid leakage during use, so B is correct. An evaporating dish can be heated directly with an alcohol lamp, and a distillation flask should be heated with wire gauze with ceramic center, so C is incorrect. An acid burette does not need to be dried before use, so D is incorrect.)

4.下列实验中, 所选装置 (可添加试剂, 可重复使用) 不合理的是 (In the following experiments, the selected device (reagents can be added and reused) is unreasonable?) ()



- A. 配制 100 mL 0.1 mol · L⁻¹ NaOH 溶液, 选用⑤(Prepare 100 mL 0.1 mol · L⁻¹ NaOH solution and select device ⑤.)
- B. 用 KMnO₄ 固体制备氧气, 选用②(Prepare oxygen with KMnO₄ solid and select device ②.)
- C. 盛放 Na₂CO₃ 溶液, 选用③(Store Na₂CO₃ solution and select device ③.)
- D. 用大理石和盐酸制取 CO₂ 并比较碳酸和 H₂SiO₃ 的酸性强弱, 选用①④(Prepare CO₂ with marble and hydrochloric acid and compare the acidity of carbonic acid and H₂SiO₃, select devices ① and ④.)

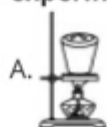
答案 (Answer):C

解析 (Analysis): 配制 100 mL 0.1 mol · L⁻¹ NaOH 溶液需选用 100 mL 容量瓶, 移液时需通过玻



璃棒引流, 则可选用⑤, A 正确; 用 KMnO_4 固体制备氧气时选用固体加热型发生装置, 故选用②, B 正确; 因为碳酸钠水解使溶液显碱性, 能和玻璃中的 SiO_2 反应得到具有黏性的 Na_2SiO_3 溶液, 实验室盛放 Na_2CO_3 溶液的试剂瓶不能用磨口玻璃塞, 即不能选用③, C 错误; 用大理石和盐酸制取 CO_2 并比较碳酸和 H_2SiO_3 的酸性强弱, 选用①可以制得 CO_2 , 但由于盐酸具有挥发性, 故需先经过盛有饱和 NaHCO_3 溶液的洗气装置后再通入盛有 Na_2SiO_3 溶液的广口瓶中, 可以选用④, D 正确。(A 100 mL volumetric flask is required to prepare 100 mL $0.1 \text{ mol} \cdot \text{L}^{-1}$ NaOH solution, and glass rod drainage is needed during liquid transfer, so device ⑤ can be selected, so A is correct. A solid heating type generator is selected to prepare oxygen with KMnO_4 solid, so device ② is selected, so B is correct. Sodium carbonate hydrolyzes to make the solution alkaline, which can react with SiO_2 in glass to form viscous Na_2SiO_3 solution. The reagent bottle for storing Na_2CO_3 solution in the laboratory cannot use a ground glass stopper, so device ③ cannot be selected, so C is incorrect. Device ① can be used to prepare CO_2 with marble and hydrochloric acid and compare the acidity of carbonic acid and H_2SiO_3 . However, due to the volatility of hydrochloric acid, it needs to pass through a gas washing device filled with saturated NaHCO_3 solution first and then into a wide-mouth bottle filled with Na_2SiO_3 solution, so device ④ can be selected, so D is correct.)

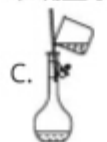
5. 下列实验操作对应的装置不正确的是 (The device corresponding to the following experimental operation is incorrect?) ()



灼烧海带制海带灰 (Burn kelp to prepare kelp ash.)



准确量取 15.00 mL 稀盐酸 (Accurately measure 15.00 mL of dilute hydrochloric acid.)



配制一定浓度的 NaCl 溶液 (Prepare NaCl solution with a certain concentration.)



使用电石和饱和食盐水制备 C_2H_2 (Prepare C_2H_2 with calcium carbide and saturated brine.)

答案 (Answer): A

解析 (Analysis): 灼烧海带制海带灰应在坩埚中进行, 给坩埚加热时不需要使用陶土网, A 错误; 稀盐酸呈酸性, 可用酸式滴定管量取 15.00 mL 稀盐酸, B 正确; 配制一定浓度的 NaCl 溶液时, 需要将烧杯中溶解得到的 NaCl 溶液通过玻璃棒引流转移到选定规格的容量瓶中, C 正确; 电石的主要成分 CaC_2 与水发生反应 $\text{CaC}_2 + 2\text{H}_2\text{O} = \text{Ca}(\text{OH})_2 + \text{CH}\equiv\text{CH}\uparrow$ 制取 C_2H_2 , 该制气反应属于固体与液体常温制气反应, 分液漏斗中盛放饱和食盐水, 具支锥形瓶中盛放电石, D 正确。(Burning kelp to prepare kelp ash should be carried out in a crucible, and wire gauze with ceramic center is not needed when heating the crucible, so A is incorrect. Dilute hydrochloric acid is acidic, and 15.00 mL of dilute hydrochloric acid can be measured with an acid burette, so B is correct. When preparing NaCl solution with a certain concentration, the NaCl solution dissolved in a beaker needs to be transferred to a volumetric flask of a specified specification through glass rod drainage, so C is correct. The main component of calcium carbide CaC_2 reacts with water: $\text{CaC}_2 + 2\text{H}_2\text{O} = \text{Ca}(\text{OH})_2 + \text{CH}\equiv\text{CH}\uparrow$ to prepare C_2H_2 . This gas production reaction is a solid-liquid gas production reaction at room temperature. Saturated brine is placed in the separatory funnel, and calcium carbide is placed in the side-arm conical flask, so D is correct.)



6.用如图所示装置进行下列实验，能达到实验目的的是 (The following experiments are carried out with the device shown in the figure, which can achieve the experimental purpose?) ()



- A. 瓶中盛有适量浓硫酸，从 A 口进气来干燥 NH_3 (An appropriate amount of concentrated sulfuric acid is contained in the bottle, and gas is introduced from the A port to dry NH_3 .)
- B. 从 B 口进气，用排空气法收集 CO_2 (Gas is introduced from the B port to collect CO_2 by air displacement method.)
- C. 瓶中装满水，从 B 口进气，用排水法收集 NO_2 (The bottle is filled with water, and gas is introduced from the B port to collect NO_2 by water displacement method.)
- D. 瓶中装满水，A 口连接导管并伸入量筒中，从 B 口进气，用排水法测量生成 H_2 的体积 (The bottle is filled with water, the A port is connected with a conduit and extended into a measuring cylinder, gas is introduced from the B port, and the volume of generated H_2 is measured by water displacement method.)

答案 (Answer): D

解析 (Analysis): 不能用浓硫酸干燥 NH_3 ，A 错误；用向上排空气法收集 CO_2 时，应从 A 口进气，B 错误； NO_2 与 H_2O 反应，不能用排水法收集，C 错误； H_2 难溶于水，从 B 口进入 H_2 ，能将瓶中水压入量筒中，进入量筒中水的体积即为 H_2 的体积，D 正确。 (Concentrated sulfuric acid cannot be used to dry NH_3 , so A is incorrect. When collecting CO_2 by upward air displacement method, gas should be introduced from the A port, so B is incorrect. NO_2 reacts with H_2O and cannot be collected by water displacement method, so C is incorrect. H_2 is slightly soluble in water. When H_2 is introduced from the B port, the water in the bottle can be pressed into the measuring cylinder, and the volume of water entering the measuring cylinder is the volume of H_2 , so D is correct.)

7.下列实验装置或操作能够达到实验目的的是 (Which of the following experimental devices or operations can achieve the experimental purpose?) ()

A	B
转移溶液 (Transfer solution.)	验证碳酸酸性强于硅酸 (Verify that carbonic acid is more acidic than silicic acid.)
C	D
除去 CO_2 中混有的少量 HCl (Remove a small amount of HCl mixed in CO_2 .)	收集 NO (Collect nitric oxide.)

答案 (Answer): A

解析 (Analysis): 转移溶液时需要用玻璃棒引流，且玻璃棒下端靠在容量瓶刻度线之下，A 正确；盐酸具有挥发性，生成的二氧化碳气体中混有氯化氢，氯化氢通入硅酸钠溶液中也有白色沉淀生成，所以无法验证碳酸的酸性强于硅酸，B 错误；二氧化碳能与饱和碳酸钠溶液反应生成碳酸氢钠，所以不能



用饱和碳酸钠溶液除去二氧化碳中混有的少量氯化氢，应用饱和碳酸氢钠溶液，C 错误；NO 易与空气中的氧气反应，不能采用排空气法收集 NO，D 错误。(Glass rod drainage is required when transferring the solution, and the lower end of the glass rod is placed below the scale line of the volumetric flask, so A is correct. Hydrochloric acid is volatile, and the generated carbon dioxide gas is mixed with hydrogen chloride. Hydrogen chloride also forms a white precipitate when introduced into sodium silicate solution, so it is impossible to verify that carbonic acid is more acidic than silicic acid, so B is incorrect. Carbon dioxide can react with saturated sodium carbonate solution to form sodium bicarbonate, so saturated sodium carbonate solution cannot be used to remove a small amount of hydrogen chloride mixed in carbon dioxide, and saturated sodium bicarbonate solution should be used, so C is incorrect. Nitric oxide easily reacts with oxygen in the air and cannot be collected by air displacement method, so D is incorrect.)

8.球形干燥管是中学化学常用的一种玻璃仪器。下列有关球形干燥管的创新用途，错误的是 (Spherical drying tube is a commonly used glass instrument in middle school chemistry. Which of the following innovative uses of spherical drying tube is incorrect?) ()



甲 ☐ ☐ ☐ 乙 ☐ ☐ ☐ ☐ ☐ ☐ 丙 ☐ ☐ ☐ ☐ 丁 ☐ ☐ ☐ ☐ ☐ ☐

- A. 甲用于尾气处理，防倒吸 (Device A is used for tail gas treatment and anti-suck back.)
 B. 乙用于铜丝与 Cl_2 反应，防污染 (Device B is used for the reaction of copper wire with Cl_2 to prevent pollution.)
 C. 丙是简易的天然水净化过滤器 (Device C is a simple natural water purification filter.)
 D. 丁用于 Na_2CO_3 固体与稀硫酸反应制备 CO_2 气体，可随制随停 (Device D is used for the reaction of Na_2CO_3 solid with dilute sulfuric acid to prepare CO_2 gas, which can be started and stopped at any time.)

答案 (Answer): D

解析 (Analysis): 球形结构可防止倒吸，甲可用于尾气处理及防倒吸，A 正确；氯气与 NaOH 反应，球形结构可防止倒吸，题图中装置可用于铜丝与 Cl_2 反应，防污染，B 正确；活性炭具有吸附性，丙装置可作简易的天然水净化过滤器，C 正确； Na_2CO_3 固体易溶于水，其与稀硫酸接触后，关闭活塞不能使固液分离，所以制备 CO_2 气体时不能随制随停，D 错误。(The spherical structure can prevent suck back, and device A can be used for tail gas treatment and anti-suck back, so A is correct. Chlorine reacts with NaOH, and the spherical structure can prevent suck back. The device in the figure can be used for the reaction of copper wire with Cl_2 to prevent pollution, so B is correct. Activated carbon has adsorption property, and device C can be used as a simple natural water purification filter, so C is correct. Na_2CO_3 solid is soluble in water. After it contacts with dilute sulfuric acid, closing the piston cannot separate solid and liquid, so CO_2 gas cannot be started and stopped at any time, so D is incorrect.)

9.化学是以实验为基础的科学，下列有关实验设计能达到目的的是 (Chemistry is an experiment-based science. Which of the following experimental designs can achieve the purpose?) ()



A	B
探究CH₄与Cl₂的取代反应(Explore the substitution reaction of CH₄ with Cl₂.)	验证氧化性: Cl₂ > Fe³⁺ > I₂ (Verify oxidizing strength: Cl₂ > Fe³⁺ > I₂.)
C	D
验证溴乙烷发生消去反应后有机产物的性质(Verify the properties of organic products after the elimination reaction of bromoethane.)	定量测定化学反应速率 (Quantitatively determine the chemical reaction rate.)

答案 (Answer): C

解析 (Analysis): 甲烷与氯气在强光作用下会爆炸, A 错误; 浓盐酸和漂白粉反应得到氯气, 氯气能将氯化亚铁氧化为氯化铁, 得到的氯化铁溶液中仍会溶解有氯气, 氯气和氯化铁都能将碘离子氧化为碘单质, 故该实验不能达到实验目的, B 错误; 溴乙烷发生消去反应得到的气体中含乙醇, 通过水后乙醇被吸收, 酸性高锰酸钾溶液可以验证乙烯的性质, C 正确; 使用长颈漏斗时装置与大气相连, 无法测定单位时间内产生气体的体积, 不能达到实验目的, D 错误。 (Methane and chlorine explode under strong light, so A is incorrect. Concentrated hydrochloric acid reacts with bleaching powder to produce chlorine, which can oxidize ferrous chloride to ferric chloride. Chlorine is still dissolved in the obtained ferric chloride solution. Both chlorine and ferric chloride can oxidize iodide ions to iodine, so this experiment cannot achieve the purpose, so B is incorrect. The gas obtained from the elimination reaction of bromoethane contains ethanol, which is absorbed after passing through water. Acidic potassium permanganate solution can verify the properties of ethylene, so C is correct. When a long neck funnel is used, the device is connected to the atmosphere, and the volume of gas generated per unit time cannot be measured, so the purpose cannot be achieved, so D is incorrect.)

10. 下列实验方法或操作叙述正确的是 (Which of the following experimental methods or operations is correct?) ()

- 用湿润的蓝色石蕊试纸检验氨气 (Test ammonia with a moist blue litmus paper.)
- 萃取实验中, 振荡后需打开分液漏斗活塞放气 (In the extraction experiment, it is necessary to open the piston of the separatory funnel to release gas after shaking.)
- 将 Na₂CO₃ 溶液保存在带玻璃塞的试剂瓶中 (Store Na₂CO₃ solution in a reagent bottle with a glass stopper.)
- 将实验用剩的金属钠块放入废物缸 (Put the remaining metal sodium block after the experiment into the waste tank.)

答案 (Answer): B

解析 (Analysis): 氨气可使湿润的红色石蕊试纸变蓝, 不能选蓝色石蕊试纸, 故 A 错误; 萃取实验中振荡操作时, 振荡后需打开分液漏斗活塞放气, 故 B 正确; 碳酸钠溶液显碱性, 会和玻璃中的 SiO₂ 反

应生成硅酸钠，使试剂瓶和玻璃塞粘在一起，不易打开，故 C 错误；金属钠性质活泼，不能直接丢弃在废物缸中，故 D 错误。

(Ammonia can turn moist red litmus paper blue, so blue litmus paper cannot be selected, so A is incorrect. During the shaking operation in the extraction experiment, it is necessary to open the piston of the separatory funnel to release gas after shaking, so B is correct. Sodium carbonate solution is alkaline and will react with SiO_2 in glass to form sodium silicate, which sticks the reagent bottle and the glass stopper together and is not easy to open, so C is incorrect. Metal sodium is active and cannot be directly discarded in the waste tank, so D is incorrect.)

11. 下列有关实验基本操作的说法错误的是 (Which of the following statements about basic experimental operations is incorrect?) ()

- A. 加热装有固体的大试管应先来回移动加热，后在药品处加热 (When heating a large test tube containing solid, it should be heated back and forth first, and then heated at the position of the medicine.)
- B. 用浓氨水清洗做过银镜反应的试管，除去银镜 (Clean the test tube after silver mirror reaction with concentrated ammonia water to remove the silver mirror.)
- C. 配制混酸时，将浓硫酸沿内壁注入装有浓硝酸的烧杯中并搅拌 (When preparing mixed acid, we should inject concentrated sulfuric acid into the beaker containing concentrated nitric acid along the inner wall and stir.)
- D. 用标准 AgNO_3 溶液滴定海水，测定其中氯离子的浓度，需要棕色滴定管、锥形瓶 (Titrate seawater with standard AgNO_3 solution to determine the concentration of chloride ions, which requires a brown burette and a conical flask.)

答案 (Answer): B

解析 (Analysis): 加热装有固体的大试管应先来回移动加热，后在药品处加热，A 正确；银与氨水不反应，应选用稀硝酸清洗做过银镜反应的试管，B 错误；配制混酸时，将浓硫酸沿内壁注入装有浓硝酸的烧杯中并搅拌，C 正确；因为 AgNO_3 溶液见光易分解，所以滴定时需要棕色滴定管、锥形瓶，D 正确。(When heating a large test tube containing solid, it should be heated back and forth first, and then heated at the position of the medicine, so A is correct. Silver does not react with ammonia water, so dilute nitric acid should be selected to clean the test tube after silver mirror reaction, so B is incorrect. When preparing mixed acid, we should inject concentrated sulfuric acid into the beaker containing concentrated nitric acid along the inner wall and stir, so C is correct. AgNO_3 solution is easy to decompose when exposed to light, so a brown burette and a conical flask are needed for titration, so D is correct.)

12. 化学基本实验操作是学好化学的基本技能之一。下列选项中操作正确的是 () (Basic chemical experimental operations are one of the basic skills to learn chemistry well. Which of the following operations is correct?)



稀释浓硫酸 (Dilute concentrated sulfuric acid)



验证质量守恒定律 (Verify the law of conservation of mass)



闻药品气味 (Smell the odor of chemicals)

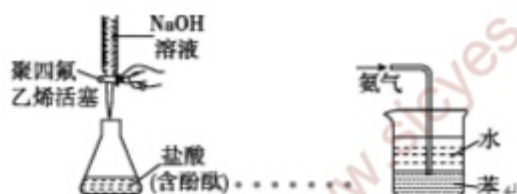


过滤液体 (Filter liquid)

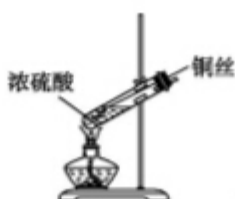
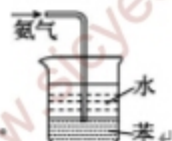
答案 (Answer): C

解析 (Analysis): 稀释浓硫酸时, 要把浓硫酸缓缓地沿器壁注入水中, 同时用玻璃棒不断搅拌, 以使热量及时扩散, 不能把水注入浓硫酸中, A 错误; $\text{Na}_2\text{CO}_3 + 2\text{HCl} = 2\text{NaCl} + \text{CO}_2\uparrow + \text{H}_2\text{O}$, 该反应有气体产生, 要验证质量守恒定律应该在密闭容器中进行, B 错误; 闻气体的气味时, 应用手在瓶口轻轻扇动, 使极少量的气体飘进鼻子中, 不能将鼻子凑到集气瓶口去闻气体的气味, C 正确; 过滤液体时, 要注意“一贴、二低、三靠”的原则, 图中缺少玻璃棒引流, D 错误。 (When diluting concentrated sulfuric acid, slowly pour concentrated sulfuric acid into water along the container wall, and stir continuously with a glass rod to dissipate heat in time; do not pour water into concentrated sulfuric acid, so A is wrong. The reaction $\text{Na}_2\text{CO}_3 + 2\text{HCl} = 2\text{NaCl} + \text{CO}_2\uparrow + \text{H}_2\text{O}$ produces gas, so the law of conservation of mass should be verified in a closed container, so B is wrong. When smelling the odor of a gas, gently fan the gas at the bottle mouth with your hand to let a very small amount of gas drift into your nose; do not put your nose close to the gas collection bottle mouth to smell the gas, so C is correct. When filtering liquid, follow the principles of one adherence, two lows, three supports a glass rod for guiding flow is missing in the figure, so D is wrong.)

13. 下图所示的实验操作及描述均正确, 且能达到相应目的的是 () (The experimental operations and descriptions shown in the figure below are all correct and can achieve the corresponding purposes.)



甲 ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ 乙



丙 ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ 丁



- A. 甲: 用于 NaOH 溶液滴定盐酸 (Figure A: Titrate hydrochloric acid with NaOH solution)
 B. 乙: 用于吸收尾气氨气 (Figure B: Absorb tail gas ammonia)
 C. 丙: 可控制浓硫酸与铜反应的发生与停止 (Figure C: Control the start and stop of the reaction between concentrated sulfuric acid and copper)
 D. 丁: 称量 10.05 g 固体 (Figure D: Weigh 10.05 g of solid)

答案 (Answer): C

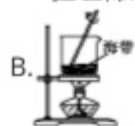
解析 (Analysis): NaOH 溶液滴定盐酸时, 应选用碱式滴定管来盛放 NaOH 溶液, 此处选用的聚四氟乙烯活塞是酸碱通用的; 该实验操作错在操作活塞的手势, A 错误。吸收氨气应注意防倒吸, 苯的密度比水小, 应在上层, 该实验操作起不到防倒吸的作用, 选用此装置应该把苯换为 CCl_4 , B 错误。通过推拉铜丝, 控制反应的发生与停止, C 正确。托盘天平的精确读数为小数点后一位, 不能读取到 0.01 g, D 错误。 (When titrating hydrochloric acid with NaOH solution, a basic burette should be used to hold NaOH solution; the polytetrafluoroethylene piston used here is universal for acid and alkali. The error lies in the gesture used to operate the piston, so A is wrong. Ammonia

absorption should prevent back suction; benzene is less dense than water and floats on the upper layer, so this device cannot prevent back suction, and benzene should be replaced with CCl_4 , so B is wrong. The start and stop of the reaction can be controlled by pushing and pulling the copper wire, so C is correct. The precision of a tray balance is one decimal place, and it cannot read 0.01 g, so D is wrong.)

14. 下列操作正确并能达到实验目的的是 () (Which of the following operations is correct and can achieve the experimental purpose?)



验证稀硝酸的还原产物为 NO (Verify that the reduction product of dilute nitric acid is NO)



灼烧海带 (Ignite kelp)



制取并收集 NH_3 (Prepare and collect NH_3)



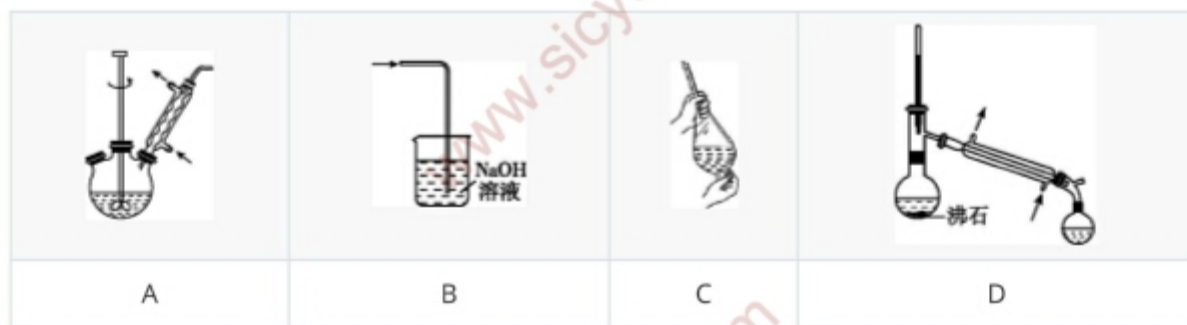
用滴定法测氨水的物质的量浓度 (Determine the amount-of-substance concentration of ammonia water by titration)

答案 (Answer): A

解析 (Analysis): 稀硝酸与碳酸钙反应生成硝酸钙、二氧化碳和水, 二氧化碳排尽装置中的空气, 再将铜丝伸入, 若无红棕色即可验证稀硝酸的还原产物为一氧化氮, A 项正确; 灼烧海带应用坩埚, 不能用烧杯, B 项错误; 氨气应利用向下排空气法收集, 则小试管中的导管应伸到试管底部, C 项错误; 盐酸与稀氨水反应生成氯化铵和水, 氯化铵溶液显酸性, 滴定终点时溶液为酸性, 而酚酞的 pH 变色范围是 $8.2 \sim 10.0$, 甲基橙的变色范围是 $3.1 \sim 4.4$, 故使用甲基橙作指示剂更加准确, D 项错误。 (Dilute nitric acid reacts with calcium carbonate to produce calcium nitrate, carbon dioxide and water. Carbon dioxide exhausts the air in the device. Then insert the copper wire; if there is no reddish-brown gas, it can be verified that the reduction product of dilute nitric acid is nitric oxide, so A is correct. A crucible should be used to ignite kelp instead of a beaker, so B is wrong. Ammonia should be collected by downward air displacement method, so the conduit in the small test tube should extend to the bottom of the test tube, so C is wrong. Hydrochloric acid reacts with dilute ammonia water to produce ammonium chloride and water. Ammonium chloride solution is acidic, and the solution is acidic at the titration end point. The pH color change range of phenolphthalein is $8.2 \sim 10.0$, and that of methyl orange is $3.1 \sim 4.4$, so methyl orange is more accurate as an indicator, so D is wrong.)

15. 某兴趣小组为制备 1-氯-2-甲基丙烷 (沸点 69°C), 将 2-甲基-1-丙醇和 POCl_3 溶于 CH_2Cl_2 中, 加热回流 (伴有 HCl 气体产生)。反应完全后倒入冰水中分解残余的 POCl_3 , 分液收集 CH_2Cl_2 层, 无水 MgSO_4 干燥, 过滤、蒸馏后得到目标产物。上述过程中涉及的装置或操作错误的是 (夹持及加热装置略) () (A research group prepared 1-chloro-2-methylpropane (boiling point 69°C) by dissolving 2-methyl-1-propanol and POCl_3 in CH_2Cl_2 and heating under reflux (accompanied by HCl gas generation). After the reaction was complete, the mixture was poured into ice water to decompose residual POCl_3 . The CH_2Cl_2 layer was collected by liquid separation, dried with anhydrous MgSO_4 , filtered and distilled to obtain the target product. Which of

the devices or operations involved in the above process is wrong (clamping and heating devices are omitted)?)



答案 (Answer): B

解析 (Analysis): 将 2 - 甲基 - 1 - 丙醇和 POCl_3 溶于盛在三口烧瓶中的 CH_2Cl_2 中, 搅拌、加热回流 (反应装置中的球形冷凝管用于回流, 冷凝水下进上出), 制备产物, A 项正确; 产生的 HCl 可用 NaOH 溶液吸收, 但要防止倒吸, 导气管不能直接插入 NaOH 溶液中, B 项错误; 分液收集 CH_2Cl_2 层, 需用到分液漏斗, 振荡摇匀时需将分液漏斗倒转过来, C 项正确; 蒸馏时需要用温度计控制温度, 冷凝水从下口进、上口出, D 项正确。 (Dissolve 2-methyl-1-propanol and POCl_3 in CH_2Cl_2 in a three-necked flask, stir and heat under reflux (the spherical condenser in the reaction device is used for reflux, and cooling water enters from the bottom and exits from the top) to prepare the product, so A is correct. The generated HCl can be absorbed by NaOH solution, but back suction should be prevented, and the gas guide tube cannot be directly inserted into NaOH solution, so B is wrong. A separatory funnel is needed to collect the CH_2Cl_2 layer by liquid separation, and the separatory funnel should be inverted when shaking evenly, so C is correct. A thermometer is needed to control the temperature during distillation, and cooling water enters from the bottom and exits from the top, so D is correct.)

16. 化学试剂不变质是化学实验成功的保障。下列试剂的保存方法不正确的是 () (Keeping chemical reagents from deteriorating is the guarantee for the success of chemical experiments. Which of the following preservation methods for reagents is incorrect?)

- A. 保存 FeCl_3 溶液时, 通常在溶液中加入少量的单质铁 (When preserving FeCl_3 solution, a small amount of elemental iron is usually added to the solution)
- B. 液溴易挥发, 在存放液溴的试剂瓶中应加水液封 (Liquid bromine is volatile, so water sealing should be applied in the reagent bottle storing liquid bromine)
- C. 实验室中金属钠通常保存在煤油里 (Metallic sodium is usually stored in kerosene in the laboratory)
- D. 浓硝酸通常保存在棕色试剂瓶中, 并置于阴凉处 (Concentrated nitric acid is usually stored in a brown reagent bottle and placed in a cool place)

答案 (Answer): A

解析 (Analysis): 铁与氯化铁反应生成氯化亚铁, 所以保存氯化铁溶液不能加入单质铁, A 错误; 液溴在水中的溶解度较小, 而且密度比水大, 所以在存放液溴的试剂瓶中应加水液封, 防止液溴挥发, B 正确; 钠能和水、氧气反应, 故应隔绝水和空气保存, 钠的密度比煤油大, 可以保存在煤油中, C 正确; 浓硝酸不稳定, 受热或见光易分解, 所以浓硝酸通常保存在棕色试剂瓶中, 置于阴凉处, D 正确。 (Iron reacts with ferric chloride to form ferrous chloride, so elemental iron cannot be added to preserve ferric chloride solution, so A is wrong. Liquid bromine has low solubility in water and is denser than water, so water sealing should be applied in the reagent bottle storing liquid bromine to prevent volatilization, so B is correct. Sodium reacts with water and oxygen, so it should be stored away from water and air; sodium is denser than kerosene and can be stored in kerosene, so C is correct. Concentrated nitric acid is unstable and easily decomposes when heated or exposed to light, so it is usually stored in a brown reagent bottle in a cool place, so D is correct.)

17. 下列试剂的保存不正确的是 () (Which of the following preservation methods for reagents is incorrect?)

选项 (Option)	化学试剂 (Chemical Reagents)	试剂瓶或用品 (Reagent Bottles or Laboratory Supplies)	保存措施 (Preservation Measures)
A	Na_2O_2 、 Na_2SO_3 、 $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$	广口瓶 (wide-mouth bottle)	密封 (sealed)
B	NaOH 溶液、新制氯水、浓 HNO_3 (NaOH solution, freshly prepared chlorine water, concentrated HNO_3)	细口配玻璃塞的玻璃瓶 (narrow-mouth glass bottle with glass stopper)	密封、避光、阴凉 (sealed, protected from light, cool)
C	汽油、苯 (Gasoline, benzene)	细口配玻璃塞的玻璃瓶 (narrow-mouth glass bottle with glass stopper)	密封、阴凉 (sealed, cool)
D	铝片、镁带 (Aluminum sheets, magnesium strips,)	塑料袋或广口瓶 (plastic bags or wide-mouth bottles)	远离强氧化剂及火源 (kept away from strong oxidants and fire sources)

答案 (Answer): B

解析 (Analysis): 三种物质均为易与空气中的水或氧气发生反应的固体, 要密封于广口瓶中保存, A 项正确; NaOH 溶液可与玻璃中的二氧化硅反应生成硅酸钠, 会使瓶塞与瓶颈粘结在一起, 不易打开, B 项错误; 汽油、苯均为易挥发的液体, 且能使橡胶塞溶胀, 需用玻璃塞密封于细口玻璃瓶中, 置于阴凉处, C 项正确; Mg 、 Al 易与空气中的氧气反应生成致密的氧化物薄膜而阻止 Mg 、 Al 进一步和氧气反应, 可保存在塑料袋或广口瓶中, 需要远离强氧化剂及火源, 以防着火, D 项正确。 (The three substances are solids that easily react with water or oxygen in the air and should be sealed and stored in wide-mouth bottles, so A is correct. NaOH solution reacts with silicon dioxide in glass to form sodium silicate, which sticks the bottle stopper to the bottle neck and makes it difficult to open, so B is wrong. Gasoline and benzene are volatile liquids that can swell rubber stoppers, so they should be sealed in narrow-mouth glass bottles with glass stoppers and placed in a cool place, so C is correct. Mg and Al easily react with oxygen in the air to form a dense oxide film to prevent further reaction with oxygen; they can be stored in plastic bags or wide-mouth bottles and kept away from strong oxidants and fire sources to prevent fire, so D is correct.)

18. 下列实验操作或处理方法错误的是 () (Which of the following experimental operations or treatment methods is wrong?)

- A. 点燃 H_2 前, 先检验其纯度 (Check the purity of H_2 before igniting it)
- B. 金属 K 着火, 用湿抹布盖灭 (Put out the fire of metallic K with a wet rag)
- C. 温度计中水银洒落地面, 用硫粉处理 (Treat spilled mercury from a thermometer with sulfur powder)
- D. 苯酚沾到皮肤上, 先后用乙醇、水冲洗 (Wash phenol on the skin with ethanol and then water)

答案 (Answer): B

解析 (Analysis): H_2 为易燃气体, 在点燃前需验纯, A 正确; 金属 K 为活泼金属, 可以与水发生反



应，不能用湿抹布盖灭，B 错误；Hg 有毒，温度计打碎后应用硫粉覆盖，使 Hg 转化为难以挥发的 HgS，C 正确；苯酚易溶于乙醇等有机溶剂，苯酚沾到皮肤上后应立即用乙醇冲洗，再用水冲洗，D 正确。（H₂ is a flammable gas and its purity must be checked before ignition, so A is correct. Metallic K is an active metal that reacts with water and cannot be put out with a wet rag, so B is wrong. Hg is toxic; sulfur powder should be used to cover broken thermometer to convert Hg into non-volatile HgS, so C is correct. Phenol is soluble in organic solvents such as ethanol; wash phenol on the skin with ethanol immediately and then with water, so D is correct.)

19. 下列实验事故的处理方法不合理的是 () (Which of the following treatment methods for experimental accidents is unreasonable?)

选项 (Option)	实验事故 (Laboratory Accidents)	处理方法 (Disposal Methods)
A	NaOH沾在皮肤上 (NaOH gets on the skin)	先用大量水冲洗,再涂上1%的硼酸 (rinse with plenty of water first, then apply 1% boric acid)
B	硫黄粉撒在实验台上 (Sulfur powder is scattered on the experimental table)	在硫黄粉表面覆盖一层水银 (cover the sulfur powder with a layer of mercury)
C	不慎吸入煤气 (Accidentally inhale gas)	立刻到室外呼吸新鲜空气或做人工呼吸 (go outdoors to breathe fresh air or perform artificial respiration immediately)
D	乙醚等有机物着火 (Ether and other organic substances catch fire)	用湿布、灭火毯覆盖可燃物 (cover combustibles with wet cloths and fire blankets)

答案 (Answer): B

解析 (Analysis): NaOH 具有强腐蚀性，NaOH 沾在皮肤上应先用大量水冲洗，再涂上酸性弱且腐蚀性弱的 1% 硼酸，A 正确；硫无毒，水银具有毒性，可以用硫覆盖处理水银 (汞)，不能用水银 (汞) 处理硫，硫黄粉撒在实验台上时直接收集处理即可，B 错误；吸入煤气 (含 CO) 会导致人体血红蛋白输氧能力减弱，故需立即到室外呼吸新鲜空气或做人工呼吸，C 正确；绝大多数有机物的密度小于水，故乙醚等有机物着火时，用湿布或灭火毯覆盖即可灭火，D 正确。（NaOH is highly corrosive; rinse NaOH on the skin with plenty of water first, then apply 1% boric acid with weak acidity and low corrosivity, so A is correct. Sulfur is non-toxic and mercury is toxic; sulfur can be used to treat mercury, but mercury cannot be used to treat sulfur. Collect and dispose of scattered sulfur powder directly, so B is wrong. Inhaling gas (containing CO) weakens the oxygen-carrying capacity of human hemoglobin, so go outdoors to breathe fresh air or perform artificial respiration immediately, so C is correct. Most organic substances are less dense than water; cover ether and other organic substances with wet cloths or fire blankets to extinguish the fire, so D is correct.)

20. 做化学实验时，必须十分重视安全和环保问题。下列操作方法不符合要求的是 () (Safety and environmental protection must be paid great attention to when conducting chemical experiments. Which of the following operation methods does not meet the requirements?)

- A. 制取氧气时，用排水法收集氧气出现倒吸现象，立即打开试管上的橡皮塞 (When preparing oxygen, back suction occurs when collecting oxygen by water displacement method, open the rubber stopper on the test tube immediately)
- B. 在气体发生装置上点燃氢气等可燃性气体时，必须先检验气体的纯度 (When igniting flammable



gases such as hydrogen on a gas generator, the purity of the gas must be checked first)

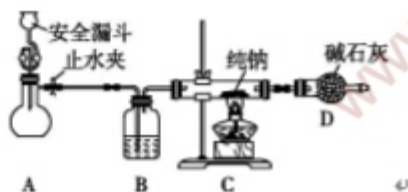
C. 实验结束后将所有的废液倒入下水道排出实验室, 以免污染实验室 (Pour all waste liquid into the sewer to discharge from the laboratory after the experiment to avoid polluting the laboratory)

D. 给试管中的液体加热时, 不停移动试管或加入碎瓷片, 以免液体暴沸伤人 (When heating liquid in a test tube, move the test tube continuously or add broken porcelain pieces to prevent liquid bumping from hurting people)

答案 (Answer): C

解析 (Analysis): 产生倒吸的原因是装置内压强减小, 使收集装置中水槽里的水进入制气装置, 打开试管上的橡皮塞可防止倒吸的水进入制气装置中, A 符合要求; 点燃可燃性气体之前, 要对气体进行验纯, 防止出现爆炸事故, B 符合要求; 实验结束后的废液中部分含有毒物质, 不能直接倒入下水道, 应倒入指定的容器中, 经处理后再排放, C 不符合要求; 给试管中的液体加热时, 不停移动试管可使试管内的液体受热均匀, 加入碎瓷片也能防止暴沸, D 符合要求。 (Back suction occurs because the pressure in the device decreases, causing water in the water tank of the collection device to enter the gas preparation device. Opening the rubber stopper on the test tube can prevent back suction water from entering the gas preparation device, so A meets the requirements. Check the purity of flammable gases before ignition to prevent explosion, so B meets the requirements. Some waste liquid after the experiment contains toxic substances and cannot be poured directly into the sewer; it should be poured into a designated container and discharged after treatment, so C does not meet the requirements. When heating liquid in a test tube, moving the test tube continuously makes the liquid heated evenly, and adding broken porcelain pieces also prevents bumping, so D meets the requirements.)

21. NaH 是有机合成中用途很广泛的物质。已知 NaH 遇水蒸气剧烈反应, 某小组设计如图所示实验装置制备 NaH, 下列说法正确的是 () (NaH is a widely used substance in organic synthesis. It is known that NaH reacts violently with water vapor. A group designed the experimental device shown in the figure to prepare NaH. Which of the following statements is correct?)

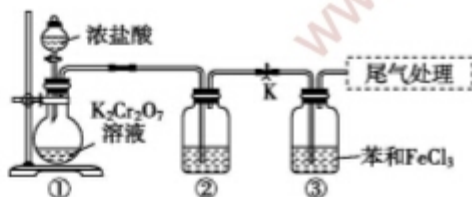


- A. 安全漏斗的作用是“液封” (The function of the safety funnel is liquid sealing)
- B. 装置 A 中的试剂是稀硝酸和粗锌 (The reagents in Device A are dilute nitric acid and crude zinc)
- C. 装置 B 中的试剂是饱和食盐水 (The reagent in Device B is saturated salt water)
- D. 实验开始后先点燃 C 处酒精灯, 再启动 A 中反应 (After the experiment starts, light the alcohol lamp at C first, then start the reaction in A)

答案 (Answer): A

解析 (Analysis): 加入液体后, 部分液体存于安全漏斗上部弯管中起液封作用, 避免氢气从安全漏斗中逸出, A 正确; 硝酸具有强氧化性, 与 Zn 反应不生成氢气, B 错误; 装置 B 是干燥氢气的装置, 装置 B 中试剂应该为浓硫酸, C 错误; 实验开始后应先启动 A 中反应, 利用生成的氢气将装置中的空气排出, 防止钠与氧气反应, 后点燃 C 处酒精灯, D 错误。 (After adding liquid, part of the liquid is stored in the upper bent tube of the safety funnel for liquid sealing to prevent hydrogen from escaping from the safety funnel, so A is correct. Nitric acid has strong oxidizing property and does not produce hydrogen when reacting with Zn, so B is wrong. Device B is a hydrogen drying device, and the reagent in Device B should be concentrated sulfuric acid, so C is wrong. After the experiment starts, start the reaction in A first to exhaust the air in the device with generated hydrogen to prevent sodium from reacting with oxygen, then light the alcohol lamp at C, so D is wrong.)

22. 氯苯是制备常用药对乙酰氨基酚的原料之一，实验室可通过苯与干燥的 Cl_2 在催化剂作用下制备氯苯，装置如图所示。下列说法错误的是 () (Chlorobenzene is one of the raw materials for preparing the common drug paracetamol. Chlorobenzene can be prepared in the laboratory from benzene and dry Cl_2 under the action of a catalyst. The device is shown in the figure. Which of the following statements is wrong?)



- A. 装置①中向圆底烧瓶中滴加浓盐酸前需先打开 K (Open K first before dropping concentrated hydrochloric acid into the round bottom flask in Device ①)
- B. 装置②的作用是吸收 HCl 气体 (The function of Device ② is to absorb HCl gas)
- C. 分离氯苯的操作为水洗、碱洗、水洗、干燥、蒸馏 (The operation for separating chlorobenzene is water washing, alkali washing, water washing, drying and distillation)
- D. 可将尾气通入氢氧化钠溶液中进行尾气处理 (Tail gas can be passed into sodium hydroxide solution for treatment)

答案 (Answer): B

解析 (Analysis): 装置①利用浓盐酸和 $\text{K}_2\text{Cr}_2\text{O}_7$ 反应制备 Cl_2 ，氯气需干燥后再参与反应， Cl_2 是酸性气体，可使用浓硫酸干燥，HCl 不会导致③中反应生成副产物，不用除去。苯和干燥的 Cl_2 在 FeCl_3 催化下反应生成氯苯和 HCl，反应后的尾气中有残留的 Cl_2 和 HCl 气体，可将尾气通入 NaOH 溶液中进行尾气处理。若关闭 K，向烧瓶中加浓盐酸，生成的 Cl_2 使烧瓶中压强增大，浓盐酸不能顺利流下，故向圆底烧瓶中滴加浓盐酸前需先打开 K，A 正确；由分析可知，装置②的作用是干燥 Cl_2 ，B 错误；分离出粗产品后，先水洗除去 FeCl_3 、HCl 等可溶性杂质，再用碳酸钠溶液碱洗，然后水洗除去多余的碳酸钠溶液，干燥后蒸馏得到氯苯，C 正确；由分析可知，可将尾气通入氢氧化钠溶液中进行尾气处理，D 正确。(Device ① prepares Cl_2 by the reaction of concentrated hydrochloric acid and $\text{K}_2\text{Cr}_2\text{O}_7$. Chlorine gas needs to be dried before reaction; Cl_2 is an acidic gas and can be dried with concentrated sulfuric acid. HCl does not cause by-products in the reaction in ③ and does not need to be removed. Benzene reacts with dry Cl_2 under the catalysis of FeCl_3 to form chlorobenzene and HCl. The tail gas after reaction contains residual Cl_2 and HCl, which can be treated by passing into NaOH solution. If K is closed, the generated Cl_2 increases the pressure in the flask when adding concentrated hydrochloric acid, making concentrated hydrochloric acid unable to flow down smoothly. Therefore, open K first before dropping concentrated hydrochloric acid into the round bottom flask, so A is correct. As analyzed, the function of Device ② is to dry Cl_2 , so B is wrong. After separating the crude product, first wash with water to remove soluble impurities such as FeCl_3 and HCl, then wash with sodium carbonate solution, then wash with water to remove excess sodium carbonate solution, dry and distill to obtain chlorobenzene, so C is correct. As analyzed, tail gas can be passed into sodium hydroxide solution for treatment, so D is correct.)